Philippe Fontaine

Data Scientist · Economist

Public policy evaluation · Fraud detection

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Profile

Data scientist at the French Senate with over ten years of experience in economic analysis and public policy evaluation. I work with large administrative datasets, simulate policy reforms and build decision-support applications. My work combines quantitative methods, public finance expertise and clear communication of findings to decision-makers.

Professional experience

Data Scientist 2025–present

French Senate | Paris

- Conduct data analysis and simulations, and prepare briefing notes for Senate committees' legislative, evaluation and government oversight activities.
- Contribute to reports on public support for businesses, mental health, construction standards and funding to combat violence against women.

Data Scientist 2021–2025

General Inspectorate of Finance (IGF) | Paris

- **Fraud detection:** developed a rules engine and a reference database of **several million prices** collected from the web to detect fraud in energy transition and international trade; built an R Shiny application enabling operational teams to reproduce the analyses independently.
- **Healthcare:** analysed **several billion healthcare claims** using the French national health data system (SNDS) and DREES statistical datasets; simulated reforms to coverage for long-term conditions (ALD) with R and Spark.
- **Decision support:** built a government revenue dashboard using Eurostat data, used by the French Budget Directorate for the **2024 spending reviews**.
- **Internal applications:** developed two Shiny applications covering HR management, assignment tracking and GDPR compliance.

Deputy Head of Unit, FIPU2 2018–2021

French Treasury | Paris

- Analysed and costed corporate tax policies; contributed to impact assessments and negotiations on international minimum taxation under the OECD's BEPS project.
- Modernised and automated the production of general government accounts.

Statistician – Investment and Tax 2015–2018

Insee (French National Statistics Office) – Quarterly National Accounts | Paris

- Oversaw quarterly estimates of investment and taxes using econometric models and short-term economic indicators.
- Produced tax revenue forecasts for Insee's economic outlook publication.

Skills

Analysis and modelling Econometrics, microsimulation, time series, predictive modelling and policy costing.

Programming and data R (data.table, Shiny, Quarto), Python (pandas, scikit-learn), SQL, Spark; web scraping and large-scale data processing.

Applications and deployment Dashboards and interactive visualisation; HTML, CSS, JavaScript; Git, Docker and GitLab CI/CD pipelines.

Domain expertise Public finance, corporate taxation, national accounts, health data and fraud detection.

Communication Policy briefs and reports; clear presentation of findings, assumptions and analytical limitations.

Personal projects – open data

Seven interactive applications developed in my own time, showcased on my website and listed on France's open data portal, data.gouv.fr. Selected examples:

- **Cantines en France**: exploration of France's national canteen register by location and management model; standardised datasets available in CSV and Parquet.
- **Organisation de l'État**: interactive directory of French ministries and public institutions using the DILA government directory, including service responsibilities and locations.
- **Obs'COP 2025**: comparison of climate change attitudes across **30 countries**, with maps and population profiles.

Portfolio and applications: philippefontaine.eu

Education

Master's degree (M2) – Public Policy Evaluation and Decision-Making 2019

ENSAI / University of Rennes

Insee Statistician Officer Programme 2013–2015

ENSAI – National School for Statistics and Information Analysis, Rennes

Preparatory classes in mathematics and physics 2011–2013

Lycée Alphonse Daudet, Nîmes

Contributions to public reports

Selection from 16 reports; titles translated and shortened. Reports in French.

- **2025.** Transparency and evaluation of public support for businesses.
- **2024.** Spending review of the long-term conditions scheme.
- **2022.** Management of nursing homes in the Orpea group.
- **2020.** International corporate tax rules: economic effects for France.

Full list and reports: philippefontaine.eu

Languages and interests

Languages French; English (professional working proficiency).

Interests Cybersecurity (TryHackMe, Hack The Box), social psychology and electronic music production.